

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	M Lakshmi Sabithi	Reg. No.:	192424280
Section / Batch:	3rd (AI & DS)	Date:	

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2,Q3	40 60 (2X30)
100 Marks			

SECTION A – APPLICATION BASED QUESTIONS

Q1. Nearest Neighbor Distance (2D Points)

Given a set of points $\{(1,2), (4,6), (7,8), (2,3), (5,5)\}$, apply the brute force approach to compute the distance between all pairs. Clearly interpret the problem and justify the method. Show all distance calculations step-by-step. Identify the nearest pair and count the number of comparisons. Analyze the time complexity of the approach. Verify the correctness of the result. Interpret its significance in spatial data analysis.

Q2. Linear Search – Analytical Problem

Given an unsorted dataset $[23, 45, 12, 67, 34, 89, 10]$, apply Linear Search to find the element 34. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required to locate the element. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the result. Interpret the efficiency of Linear Search in real-world scenarios

Code of Conduct

-192424290

I M. habshmi Sahithi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Sahithi

RUBRICS FOR CALCULATION

Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured,	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach

Nearest Neighbour Distance (Brute Force)

given points,

A(1,2), B(4,6), C(7,8), D(2,3), E(5,5)

Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Step 1: Calculate all pair distances

Pair	Distance Calculation	Distance
A-B	$\sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16}$	5.00
A-C	$\sqrt{(7-1)^2 + (8-2)^2} = \sqrt{72}$	8.49
A-D	$\sqrt{(2-1)^2 + (3-2)^2} = \sqrt{2}$	1.41
A-E	$\sqrt{(5-1)^2 + (5-2)^2} = \sqrt{25}$	5.00
B-C	$\sqrt{(7-4)^2 + (8-6)^2} = \sqrt{13}$	3.61
B-D	$\sqrt{(2-4)^2 + (3-6)^2} = \sqrt{13}$	3.61
B-E	$\sqrt{(5-4)^2 + (5-6)^2} = \sqrt{2}$	1.41
C-D	$\sqrt{(2-7)^2 + (3-8)^2} = \sqrt{50}$	7.07
C-E	$\sqrt{(5-7)^2 + (5-8)^2} = \sqrt{13}$	3.61
D-E	$\sqrt{(5-2)^2 + (5-3)^2} = \sqrt{13}$	3.61

Step 2: Nearest pair

Minimum Distance = 1.41

Nearest pairs are:

A(1,2) and D(2,3)

B(4,6) and E(6,5)

Step 3: Number of comparisons

For $n=5$ points,

$$\frac{n(n-1)}{2} = \frac{5 \times 4}{2} = 10$$

comparisons = 10

Step 4: Time Complexity

Brute force: $O(n^2)$

Step 5: Verification

The smallest calculated distance is 1.41, so the result is correct.

Step 6: Significance

Finds the closest pair of points accurately.

Used in GIS, robotics, GPS navigation, image processing, spatial data analysis.

Linear Search - An

Array : [23, 45, 12, 67, 34, 89, 10]

key = 34

Step-by-step Search

Comparison	Element	Result
1	23	Not found
2	45	Not found
3	12	Not found
4	67	Not found
5	34	found

Number of comparisons = 5

Best, Average and Worst case

Best case: 1 comparison $\rightarrow O(1)$

Average case: $(n+1)/2$ comparisons ≈ 4 comparisons $\rightarrow O(n)$

Worst case: 7 comparisons $\rightarrow O(n)$

Verification:

Element 34 is found at the 5th position (index 4)

Time Complexity

- + Best case = $O(1)$
- + Average case = $O(n)$
- + Worst case = $O(n)$

Real-World Applications

- + Searching in small or unsorted datasets.
- + Contact lists, attendance records, and inventory lists.
- + Preferred when sorting is not possible or the dataset is small.